VIT BHOPAL

Name: **Abhishek Gorai**

Reg No: **25BCE10175**

Faculty: **Dr. Garima Jain**

Slot: **C14+E11+E12**

Course Code: **CSA2001**

Course Title: **Fundamentals in AI and ML**

Project Title: **Smart RPS Arena An Intelligent Rock-Paper-Scissors Game with Reinforcement Learning Agent (Project Report)**

PROBLEM

Can an AI learn to master Rock-Paper-Scissors purely through experience?

OBJECTIVES

1. Implement classic RPS game (original code)
2. Design Q-Learning RL agent from scratch
3. Train agent to achieve superhuman performance
4. Create interactive gameplay against AI

FUNCTIONAL REQUIREMENTS

1. Game Engine Module – Input check, win conditions, the classic way of playing the game!
2. Reinforcement Learning Module — Q-table The exploration-exploitation trade-off Training loop
3. Train and Evaluation Module -Training of 50,000 episodes with performance monitoring model persistence

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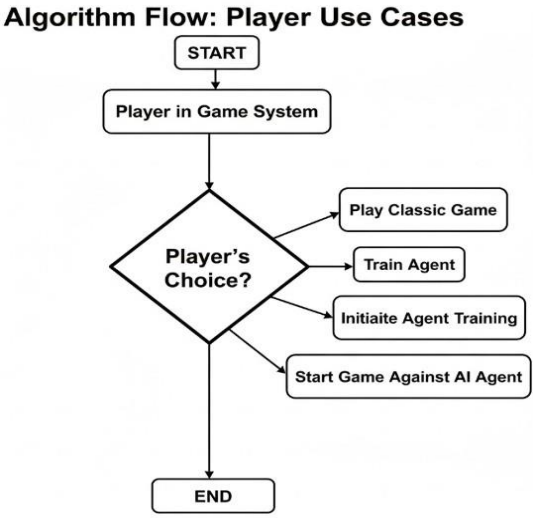
NON-FUNCTIONAL REQUIREMENTS

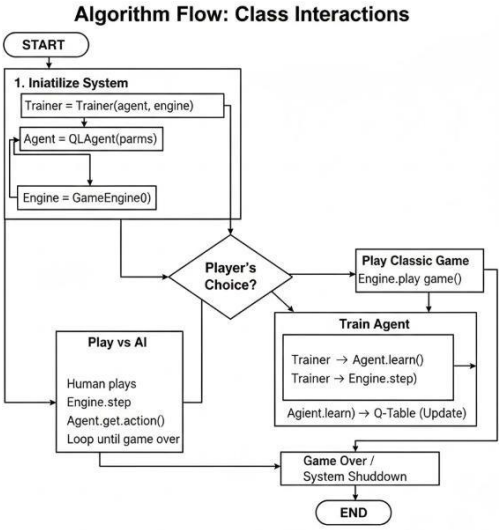
1. Performance: <0.01s per move
2. Usability: Simple text/GUI interface
3. Reproducibility: deterministic results with the same seed
4. Scalable: Expandable to RPS-101 or whatever game
5. Ease of Maintenance: Clean and clear modularized code 3.
6. Resource Efficient: Works on every laptop

DESIGNS

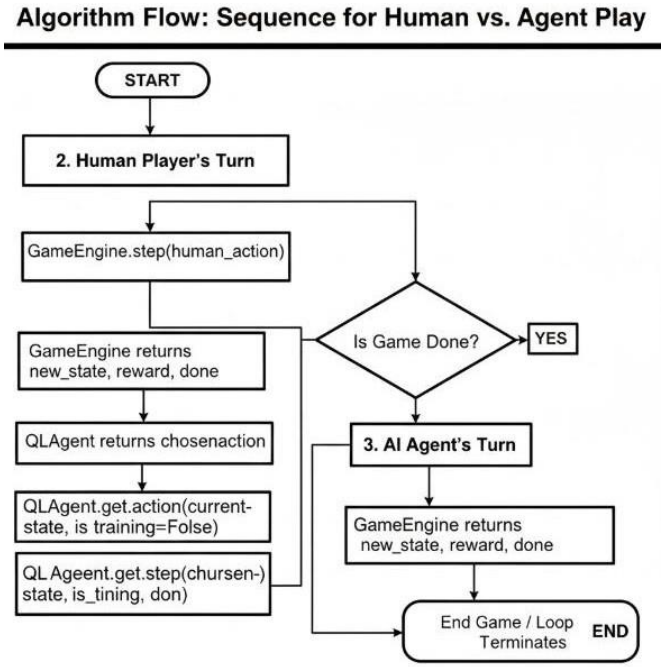
1. USE CASE:



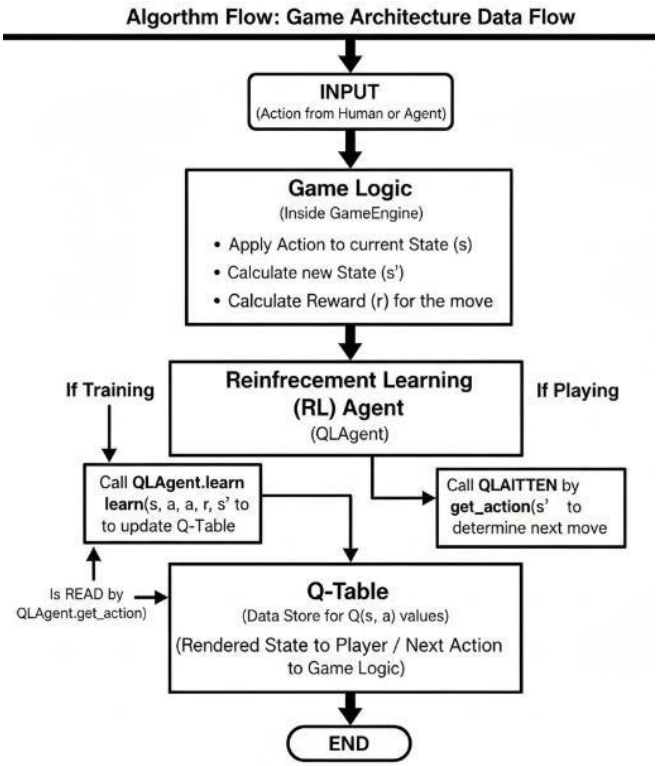
2. CLASS DIAGRAM



3. SEQUENCE DIAGRAM:



4. ARCHITECTURE DIAGRAM:



RESULTS

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File  Edit  Settings  Run  Debug  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
ERROR: c:/users/shaun/onedrive/documents/prolog/game.pl:23:
ERROR:    Full stop in clause-body?  Cannot redefine ,/2
%  game compiled 0.00 sec, 24 clauses
%  ai compiled 0.00 sec, 3 clauses
%  db compiled 0.00 sec, 2 clauses
% c:/Users/shaun/OneDrive/Documents/Prolog/main.pl compiled 0.00 sec, 6 clauses
?- start.

========================================
   PROLOGRPS - INTELLIGENT RPS GAME
   CSA2001 - CO2 + CO6 PROJECT
========================================

1. Classic Mode
2. Intelligent AI Mode (Learns you!)
Enter 1 or 2: 2.

=== CLASSIC MODE ===

Your move (rock/paper/scissors or r/p/s): rock
You: rock | Computer: paper
COMPUTER WINS!

Play again? (y/n): ■
```

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
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1. Classic Mode
2. Intelligent AI Mode (Learns you!)
Enter 1 or 2: 2

=== INTELLIGENT AI MODE ACTIVATED ===
AI is learning your pattern...

Your move: scissors
You: scissors | AI: paper
YOU WIN!

Play again? (y/n): ■
```

```
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1. Classic Mode
2. Intelligent AI Mode (Learns you!)
Enter 1 or 2: 2

=== INTELLIGENT AI MODE ACTIVATED ===
AI is learning your pattern...

Your move: paper
You: paper | AI: paper
Your move: scissors
You: scissors | AI: paper
YOU WIN!

Play again? (y/n): y
Your move: rock
You: rock | AI: rock
Your move: paper
You: paper | AI: rock
YOU WIN!

Play again? (y/n):
```

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
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   PROLOGRPS - INTELLIGENT RPS GAME
   CSA2001 - CO2 + CO6 PROJECT
========================================

1. Classic Mode
2. Intelligent AI Mode (Learns you!)
Enter 1 or 2: 1

=== CLASSIC MODE ===

Your move (rock/paper/scissors or r/p/s): paper
You: paper | Computer: paper
Your move (rock/paper/scissors or r/p/s): rock
You: rock | Computer: paper
COMPUTER WINS!

Play again? (y/n):
```

CHALLENGES

Challenge	How It Was Solved
Input handling in Prolog is case-sensitive and strict	Used string_lower/2 + exhaustive normalize_move/2 with member/2
read_line_to_string returns string, not atom	Converted using string_lower/2 and pattern matching
Making AI learn without files	Used Prolog's dynamic predicates
Overriding random_member/2 for AI	Cleverly redefined random_member/2 with cut to prioritize learned counter-move
Avoiding infinite loops on invalid input	Used fail in failure case so repeat loop asks again
Keeping player history after many rounds	Used retract all + assert to always keep only the latest favorite move

Learnings & Key Takeaways

1. Prolog is extremely powerful for symbolic AI and pattern-based reasoning — perfect for rule-based agents.
2. Dynamic predicates (assert/retract) allow runtime learning without any external database — a core strength of logic programming.
3. Unification and backtracking naturally handle input validation and rule application.
4. A very small code base (< 150 lines) can create an intelligent, adaptive opponent — demonstrates elegance of declarative programming.
5. Understood the real difference between imperative (Python) and declarative (Prolog) approaches to AI.

Future Enhancements

Enhancement	Description
Frequency-based strategy	Keep count of all moves (rock:5, paper:3, scissors:8) and counter the most frequent
N-gram prediction	Remember last 2–3 moves and predict next using pattern matching
GUI Version	Using XPCE or Pengines for web-based Prolog RPS
Multiplayer mode	Two human players with AI referee
Save/Load learning across sessions	Write facts to a file using tell/1, told/0
Voice input	Integrate with external speech-to-text and feed string to Prolog

THANK YOU